UV Mask Maker V4 User Manual (English)

This is the user manual for **UV Mask Maker V4**, an intuitive Unity editor extension designed to create, edit, and apply mask textures directly on 3D models in Unity.

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1. Tool Overview

UV Mask Maker V4 is a Unity editor extension that allows you to select polygons/UVs and paint directly on 3D meshes (`SkinnedMeshRenderer` / `MeshRenderer`)—such as VRChat avatars and costumes—within the Unity SceneView to generate, export, and apply black-and-white mask textures.

Without round-tripping between external painting applications (Photoshop, Clip Studio Paint, etc.) and Unity, you can preview changes directly on the model in Unity and generate/assign masks in just a few simple steps.

Key Highlights

- **Zero-Travel Workflow**: No need to move your mouse back and forth to screen edges. Operations are completed under your hand via the Radial Menu under the cursor (`Hold C`) and in-place brush resizing (`Ctrl + Right-Drag`).
- **Smart Toggle & Intuitive Selection**: No separate "Add" and "Remove" modes. Clicking an unselected region adds it, while clicking a selected region removes it. Double-clicking instantly selects an entire connected garment piece or accessory.
- **Symmetry Editing (X-Mirror)**: Automatically identifies symmetrical mesh counterparts across the X-axis. Selections and brush strokes on arms, legs, or shoes are instantly mirrored to the other side, cutting operations in half.
- **One-Click Start & Auto-Recovery**: Selecting an object and clicking "Start" automatically handles mesh analysis, isolation, camera framing, and session recovery.
- **Quick Export & Smart Apply**: Press `Ctrl + Enter` to export instantly. Click the **[Apply]** button on the toast notification to assign the generated mask directly into shader mask slots (lilToon, Poiyomi, etc.).
- **UV Wireframe Export**: Export clean UV wireframe PNGs with one click to use as reference backdrops in external painting software.

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2. Quick Start (Minimal Usage)

Here is the quickest workflow to select part of a costume, export a mask, and apply it to a material:

```
[Select in Hierarchy] → [Right-Click to Start] → [Click Mesh to Select] →
[Ctrl+Enter to Export] → [Click [Apply]]
```

Step 1: Start Editing

1. In the **Hierarchy**, select the object (mesh of an avatar or outfit) you want to create a mask for.
2. Right-click and choose **`dennokoworks` > `MaskMaker V4`**.

- *(Alternatively, open the window from the top menu `dennokoworks` > `MaskMaker V4` and click **"Start"**).*

3. The target mesh is automatically isolated, and the SceneView camera frames it.

Step 2: Select Regions to Mask

1. In the **SceneView**, **Left-Click** the part you want to mask.
   - The clicked UV island is selected and displayed with a mask overlay.
   - Clicking an already selected island unselects it.
   - **Double-Click** to select the entire connected part (e.g., ribbon, shoe, button).
2. For symmetrical parts like sleeves or gloves, press `X` to turn on X-Mirror. Clicking one side will automatically select the matching opposite side.

Step 3: Export and Apply to Material

1. Press `Ctrl + Enter` on your keyboard (or click the "Export" button on the UI).
2. Once exported, a toast notification appears beside your cursor: `[SaveAs] (FileName).png [Apply]`.
3. Click the `[Apply]` button inside the toast.
   - The texture is automatically assigned to the material's mask slot (e.g., lilToon's 2nd Color Mask or Poiyomi's Decal Mask).
   - If multiple mask slots are available, a quick candidate list appears for you to pick the desired slot.

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3. Feature List

Category	Feature	Description
Selection System	**Smart Toggle**	Click unselected to add, click selected to remove (no mode switching needed)
	Stroke Selection	Left-drag to continuously select or deselect multiple islands
	Part Selection	Double-click to select an entire physically connected mesh component
	Selection Granularity	Switch between "Island" (UV island), "Part" (connected mesh), and "Face" (triangle polygon)
	Grow / Shrink	`Shift + Mouse Wheel` expands or shrinks selection to adjacent polygons
	Symmetry (X-Mirror)	Press `X` to toggle. Automatically mirrors selections across the X-axis
	Select All / Invert / Clear	Standard shortcuts: `Ctrl + A` (Select All), `Ctrl + I` (Invert), and Clear
	Hover Outline Preview	White outline previews addition; red outline previews removal before clicking
Painting System	**4 Paint Tools**	Brush, Rectangle fill, Lasso fill, and Eraser

Category	Feature	Description
	In-Place Brush Resizing	`Ctrl + Right-Drag` horizontally to smoothly adjust brush radius under the cursor
	Instant Eraser	Hold `Ctrl` while painting with any tool to temporarily erase
	Ultra-HD Internal Buffer	Hand-painted strokes are stored internally at 4096×4096 px with zero quality degradation
View & Preview	**SceneView Overlay**	Real-time mask projection and UV seam edge visualization on 3D meshes
	2D UV View	Integrated 2D UV layout editor supporting direct clicking, dragging, and painting
	Base Texture Overlay	Display the model's base texture underneath the UV view for accurate alignment
	Target Isolation	Temporarily hides unrelated scene objects while working
	X-Ray View	View and select through occluded backfaces and inner geometry
	Pause Picking	Hold `Space` to pause tool picking and use normal Unity camera/transform tools
Export & Apply	**B&W Mask PNG Export**	Exports linear/non-sRGB PNG with appropriate mipmap settings
	Padding / Dilation	Automatically dilates mask edges into empty UV space to prevent mipmap seam bleeding
	Inverted Mask Export	Export an inverted mask or generate both normal and inverted masks simultaneously
	Channel Packing	Inject the mask into a specific RGBA channel of an existing texture
	In-Place Overwrite	Re-exporting overwrites the previous file, keeping existing material references intact
	Smart Material Apply	1-click assignment to shader mask slots directly from the export toast
	UV Wireframe Export	Export UV wireframe images as reference templates for external paint programs
	Import Mask Image	Drag and drop an existing mask image into the UV View to import it into the paint layer
UI & Workflow	**Radial Pie Menu**	Hold `C` to summon a circular tool menu directly beneath the cursor
	Floating Capsule Toolbar	Compact, dockable toolbar floating inside the SceneView

Category	Feature	Description
	Unified Undo / Redo	Selection, painting, and material application share a unified history stack (`Ctrl+Z` / `Ctrl+Y`)
	Auto-Save & Restore	Automatically saves edit states; work persists across recompilations and Unity restarts

4. Detailed Guide

4.1 Selection Mode in Detail

Selection mode (switched with `Tab`) is the primary mode for masking regions along the polygon structure of the mesh.

3 Selection Granularities

Switch using keys `1`, `2`, `3` or via the Radial Menu / Capsule Toolbar:

- **1: Island (UV Island)** *(Default)*: Selects by UV island boundary. Ideal for selecting specific fabric patterns, sleeves, or accessory sections.
- **2: Part (Connected Mesh)**: Selects the entire connected geometric component sharing vertices (equivalent to double-clicking).
- **3: Face (Polygon Triangle)**: Selects individual triangle polygons. Ideal for trimming or carving out tiny sections from an island.

> [!TIP] **Switching granularity preserves existing selections**
> Fine selections made in "Face" mode are not lost when switching to "Island" mode. The granularity setting only dictates *the size of the unit for your next click*.

Mouse & Keyboard Techniques

- **Smart Toggle**: Clicking unselected geometry adds it; clicking selected geometry removes it.
- **Stroke Selection (Left-Drag)**: Click and drag across multiple islands to continuously select or deselect them in one fluid motion.
- **Forced Add / Forced Remove**:
  - `Shift + Click / Drag`: Forces **addition**, regardless of whether the target was already selected.
  - `Ctrl + Click / Drag`: Forces **removal / deselection**.
- **Grow / Shrink Selection**:
  - `Shift + Mouse Wheel Up`: Expands the selection boundary outward by one adjacent polygon step.
  - `Shift + Mouse Wheel Down`: Shrinks the selection boundary inward by one step.

X-Mirror (Symmetry)

- Press `X` to toggle symmetry.
- When enabled, the corresponding geometry across the avatar's center plane (X=0) is highlighted and updated simultaneously.

- Cuts camera navigation and click counts in half when working on symmetrical clothing, sleeves, gloves, or shoes.

4.2 Paint Mode in Detail

Press `Tab` to enter Paint Mode. Use this when you want freehand control across or within polygons.

4 Paint Tools

Quickly switch tools using the Radial Menu (`Hold C`) or hotkeys:

1. **Brush (`B`)**: Projects a circular brush onto the 3D surface.
2. **Rectangle (`M`)**: Fills a rectangular screen-space region in one drag.
3. **Lasso (`L`)**: Fills a freehand lasso region.
4. **Eraser (`Shift + B`)**: Erases painted mask strokes.

In-Place Brush Resizing (`Ctrl + Right-Drag`)

- Hold `Ctrl` **and drag the Right Mouse Button horizontally** over the mesh to smoothly resize the brush circle right under your cursor.
- Eliminates mouse travel to top sliders.
- You can also tap `[` (decrease) and `]` (increase).

Instant Eraser Modifier

- In any paint tool, holding `Ctrl` **while painting acts as an eraser**. You do not need to switch tools back and forth.

Ultra-HD Internal Buffer

- All hand-painted strokes are preserved internally at **4096×4096 px**.
- Changing the export resolution preview to 512 or 1024 will display the resulting pixelated look in real-time, but does not degrade the underlying stroke data. You can return to 2048 or 4096 at any time with crisp fidelity.

4.3 Working with the 2D UV View

The **UV View** in the central section of the tool tab allows working directly on the unwrapped 2D UV layout.

- **Unified Interaction**: The same selection tools, brush tools, and hotkeys function identically on the 2D UV canvas as in the 3D SceneView.
- **Navigation Controls**:
  - Mouse Wheel: Zoom in / Zoom out
  - Middle Mouse Drag: Pan across canvas
  - Frame Button (top-right zoom icon or `F` key): Reset view to center and 100% zoom
- **Base Texture Overlay**:
  - Click the texture icon in the UV View's top-right corner to display the model's base texture as a semi-transparent backdrop, making it easy to align masks with existing texture art.

- **Importing Existing Masks**:
  - **Drag and drop** any black-and-white texture file from your Project window into the UV View. Dark areas will be imported directly into the hand-painted mask layer (Undo supported).

4.4 Export Settings & Material Application

When ready, click "Export" or press `Ctrl + Enter` to write out the mask texture.

Primary Export Options

- **File Name**: Defaults to `{RendererName}_mask.png`.
- **Resolution**: Choose from `512`, `1024`, `2048`, or `4096` (automatically detected from the base texture).
- **Channel Packing (Optional)**:
  - When checked, writes the mask specifically into the chosen channel (R, G, B, or A) of an existing texture (Base PNG).
  - Perfect for metallic, smoothness, or ambient occlusion composite maps.

Advanced Options (Expand ▸ Details)

- **Invert Mask**: Inverts black and white values.
- **Pixel Margin (Padding)**: Dilates mask boundaries outward into unused UV space. This prevents dark or discolored mipmap bleeding along UV seams in-game.
- **Save Inverted Too**: Generates both the standard mask (`_mask.png`) and the inverted mask (`_mask_inv.png`) simultaneously.

Overwriting & Smart Material Assignment

- **Safe Overwriting**: Re-exporting from the same target mesh overwrites the previously exported file without creating cluttering numbered files (`_mask 1.png`, etc.). Existing material links update automatically.
- **Smart Apply Button**:
  - The toast notification displaying `[SaveAs] (FileName).png [Apply]` allows instant assignment to your shader.
  - It inspects popular avatar shaders (lilToon, Poiyomi, UTS2, Standard) and assigns the texture to the appropriate mask property (e.g., Decal Mask, 2nd Main Color Mask).
  - Selected slot preferences are remembered per mesh.

4.5 UV Wireframe Export

Export clean wireframe layout images for use as template layers in Clip Studio Paint, Photoshop, or other external software.

1. Click the `[Camera]` **icon button** in the top right of the UV View.
2. The "UV Layout Export" dialog will appear.
3. Configure the output:
   - **Resolution**: 512 to 4096 (Default: 2048).
   - **Elements**:

        - Polygon Wireframe (internal triangle edges).
        - Seam Lines (outer UV boundary edges).
        - Selection Only (renders only currently selected islands/polygons).
        - Island Coloring (fills each UV island with a distinct pastel color).
    - **Background**: Transparent or Base Texture.
    - **Colors**: Customize wireframe, seam, and island fill colors.
4. Click `[Export]` to save `{FileName}_uv.png` into your project.

4.6 Interface & Shortcut Reference

UI Layout

1. **Tool Tab (EditorWindow)**:
    - Opened via `dennokoworks` > `MaskMaker V4`.
    - Contains Target Mesh picker, UV View, Export settings, and the primary action button.
2. **Floating Capsule Toolbar**:
    - A rounded floating bar appearing inside the SceneView during editing. Drag to reposition anywhere.
    - Houses Mode toggle, Granularity, Paint Tools, X-Mirror, Isolation, Undo/Redo, Export, and Selection counter.
3. **Cursor Radial Menu**:
    - Hold `C` anywhere in the SceneView or UV View to open a circular menu centered directly under the cursor.
    - Flick the cursor in the direction of the desired tool and release `C` to select instantly.

Shortcut Reference Table

Key	Action
`Tab`	Toggle Selection Mode ⇄ Paint Mode
`Left Click`	Smart Toggle (Select if unselected, Deselect if selected)
`Double Click`	Select / Deselect entire connected component
`Left Drag`	Continuous stroke selection / painting
`Shift` + **Click / Drag**	Force Add (Force Paint in Paint Mode)
`Ctrl` + **Click / Drag**	Force Remove (Temporary Eraser in Paint Mode)
`Ctrl` + **Right-Drag**	In-place smooth brush resizing
`Shift` + **Mouse Wheel**	Grow (Up) / Shrink (Down) selection
`X`	Toggle Symmetry (X-Mirror)
`1` / `2` / `3`	Selection Granularity (1: Island / 2: Part / 3: Face)
`B` / `M` / `L` / `Shift+B`	Paint Tools (B: Brush / M: Rect / L: Lasso / Shift+B: Eraser)
`[` / `]`	Decrease / Increase brush size

Key	Action
`C` **(Hold)**	Open Radial Menu under cursor
`Space` **(Hold)**	Pause picking (enables standard Unity camera and transform tools)
`Ctrl + A`	Select All
`Ctrl + I`	Invert Selection
`Ctrl + Z` / `Ctrl + Y`	Unified Undo / Redo
`Ctrl + Enter`	Quick Export with current settings
`F`	Reset UV View zoom and pan to center